Community Detection In Networks: Evaluating SCORE Through Simulation & Real-World Applications

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Abstract

Community detection aims to recover latent groups from network connections, but degree heterogeneity can obscure those groups: nodes in the same community may differ substantially in their number of links. This report studies SCORE (Spectral Clustering on Ratios-of-Eigenvectors), proposed by Jin (2015) for the degree-corrected stochastic block model. SCORE clusters ratios of leading adjacency-matrix eigenvector entries, which can approximately cancel a shared node-specific degree effect. On the political blogs network, the current analysis reproduces 58 SCORE classification errors among 1,222 blogs (4.7%; ARI 0.819), compared with 437 errors for ordinary spectral clustering and 590 for normalized spectral clustering. A 50-seed check for nPCA yielded between 590 and 600 errors, indicating modest sensitivity to k-means initialization. In a preliminary DCBM simulation with 500 nodes, two communities, and degree heterogeneity fixed at sigma = 0.55, all three methods performed well across 30 networks; nPCA had the lowest mean error rate (0.007) and highest mean ARI (0.972). These contrasting results show that relative method performance depends on the network conditions and motivate further simulations that vary degree heterogeneity and community separation.

Introduction

Many datasets are naturally represented as networks. A network consists of nodes, which represent the entities being studied, and edges, which represent observed relationships between pairs of entities. In community detection, the task is to infer groups from the pattern of connections: nodes in the same group are expected to connect more often to one another than to nodes in other groups. The labels are unknown during estimation, so the method must recover the groups from the network itself.

The political blogosphere surrounding the 2004 United States presidential election gives a concrete example. Each node represents a political blog, and an edge represents a hyperlink between two blogs. Following Jin (2015), hyperlink direction is ignored, so the network is treated as undirected. The data contain two recognizable communities associated with liberal and conservative political orientations. Those orientations are useful as reference labels for evaluating a clustering method, but the method estimates groups from hyperlinks rather than being given the labels.

The challenge is that blogs differ in popularity. A widely read blog may receive or make many links, while a less prominent blog may have few. This difference in degree can occur even when two blogs belong to the same political community. A basic stochastic block model (SBM) represents connection patterns through community membership: blogs assigned to the same community share the same expected connection profile. This is a useful simplification, but it does not represent the large within-community differences in popularity that can occur in real networks. When the model is a poor fit, conventional clustering may group nodes partly by their overall number of connections rather than by the community pattern we want to identify.

The degree-corrected block model (DCBM) extends this structure by allowing each node to have its own degree tendency. It can therefore represent both the community-level pattern of connections and the fact that some nodes are hubs while others are peripheral. Estimating every node's degree tendency directly and separating it from community structure can be difficult. SCORE, introduced by Jin (2015), offers a spectral approach: it computes leading eigenvectors of the adjacency matrix and clusters ratios of their entries. The intuition is that degree heterogeneity affects these eigenvectors in a shared way, so taking ratios can largely cancel the popularity effect and leave features that more clearly distinguish communities. This cancellation is approximate, but it gives SCORE a way to address degree variation without first estimating a separate degree parameter for every node.

Compared with ordinary spectral clustering, which clusters the eigenvector coordinates directly, SCORE's ratios are designed to be less sensitive to degree heterogeneity. Normalized spectral clustering is another common strategy: it adjusts the adjacency matrix using node degrees before computing eigenvectors. Comparing these methods on the same network helps show the practical difference between using raw eigenvectors, degree normalization, and eigenvector ratios. Jin's paper develops theoretical conditions for reliable community recovery and demonstrates SCORE on the political blogs network. In this project, the blog labels provide a benchmark for comparing SCORE with ordinary and normalized spectral clustering. Synthetic DCBM experiments complement the real-data example by varying degree heterogeneity and community separation under controlled conditions. Together, these analyses ask when accounting for node popularity improves community detection and how SCORE performs relative to traditional spectral alternatives.

Methods

Network and degree-corrected block model

Let an undirected network have n nodes and adjacency matrix $A \in \{0,1\}^{n\times n}$. For distinct nodes i and j, $A_{ij} = 1$ if there is an edge and $A_{ij} = 0$ otherwise. The adjacency matrix is symmetric and has a zero diagonal when self-loops are excluded. Let $g_i \in \{1,\dots,K\}$ be the unknown community of node i, and let $\theta_i > 0$ describe its relative tendency to form edges. The DCBM assumes

$$\Pr(A_{ij} = 1 \mid g, \theta) = \theta_i \theta_j P_{g_i g_j}, \qquad i < j,$$

where P is a symmetric $K \times K$ matrix of baseline connection probabilities between communities. Conditional on the parameters, edge indicators for distinct unordered pairs are independent Bernoulli variables. The values of (P) describe the community connection pattern; the node parameters θ_i allow nodes in the same community to have different expected degrees. All probabilities must be in ([0,1]).

The community structure is low-dimensional when K is small relative to n: connection patterns are represented by the $K \times K$ matrix P, together with one community label per node. The labels are the primary quantities of interest. The degree parameters account for popularity differences that could otherwise obscure those labels.

SCORE estimation and intuition

Eigenvectors summarize large-scale patterns in a matrix. When a network has stronger connections within communities than between them, leading eigenvectors of its adjacency matrix can give nodes with similar community roles similar coordinates. Clustering those coordinates can reveal groups.

Degree heterogeneity can distort these coordinates. A node's eigenvector entries may reflect both its community and its overall tendency to form edges. Under the DCBM, the node-specific degree effect can appear as a shared multiplicative factor across eigenvector entries. SCORE uses ratios to approximately cancel this common factor.

Let $\hat{\eta}_1, \ldots, \hat{\eta}_K$ be the eigenvectors of the observed adjacency matrix A associated with the K largest eigenvalues in absolute value. SCORE constructs

$$\widehat{R}(i,k) = \frac{\widehat{\eta}_{k+1}(i)}{\widehat{\eta}_1(i)}, \qquad i = 1, \ldots, n, \quad k = 1, \ldots, K-1.$$

Each row of $\widehat{R}$ is a feature vector for one node. For two communities it has a single ratio coordinate. SCORE clips extreme coordinates to limit the effect of near-zero denominators, then applies K-means to the ratio vectors. The cancellation is approximate for an observed, noisy network; it does not guarantee perfect recovery.

Comparison methods and evaluation

The analysis compares three methods:

1. **SCORE:** K-means on ratios of leading eigenvectors of A.
2. **Ordinary spectral clustering (oPCA):** K-means on leading eigenvectors of A directly.
3. **Normalized spectral clustering (nPCA):** K-means on leading eigenvectors of $D^{-1/2}AD^{-1/2}$, where D is the diagonal matrix of node degrees.

The methods differ in how degree variation enters the clustering features: oPCA uses the raw adjacency eigenvectors, nPCA normalizes by degree, and SCORE forms ratios intended to cancel a shared degree effect.

Because cluster labels are arbitrary, the estimated label 1 is not inherently liberal or conservative. Misclassification rate is therefore minimized over the two possible label mappings. The adjusted Rand index (ARI) is a complementary measure of agreement between two partitions, adjusted for chance; larger values indicate stronger agreement, with 1 representing a perfect match.

Numerical Experiments

Political blogs network and processing

The real-data analysis uses the `blogosphere` dataset distributed with the R package `gsbm`. It contains 1,222 blogs, 16,714 undirected connections, blog names, and manually assigned political orientations. The adjacency matrix is symmetrized and self-loops are removed. The labels are held out during clustering and used only for evaluation. The code records the random seed and calculates misclassification rate and ARI for each method.

Preliminary real-data results

The separate analysis file `11_RMD_PoliticalBlogsReplication.Rmd` contains the code that fits and evaluates the methods. The current reported results are:

Method	Misclassified blogs	Error rate	ARI
SCORE	58 / 1,222	4.75%	0.819
Ordinary spectral clustering (oPCA)	437 / 1,222	35.76%	0.080
Normalized spectral clustering (nPCA)	590 / 1,222	48.28%	0.000*

The nPCA ARI is 0.000 to the displayed precision.

SCORE has the strongest agreement with the reference political labels in this run: it makes 58 errors and has ARI 0.819. This matches Jin's reported SCORE error count of 58/1,222 and the ARI recorded in the project materials. oPCA makes 437 errors (35.8%) and has ARI 0.080, indicating weak agreement. nPCA makes 590 errors (48.3%) and has ARI 0.000 to the displayed precision, indicating little adjusted agreement with the reference partition in this run. These results suggest that SCORE is the most effective of the three methods on this particular real network; they do not establish that it will always outperform degree-normalized clustering.

The nPCA result varies modestly with the k-means random seed. Across 50 seeds, the minimum, first quartile, median, and third quartile were all 590 errors; the maximum was 600, and the mean error rate was 0.4846. The earlier 600-error result is therefore consistent with a less favorable k-means initialization under this implementation. The main comparison uses a fixed seed for reproducibility, while the seed check shows the range of initialization sensitivity.

The SCORE ratio plot is consistent with the table: label 0 blogs tend to have higher ratio values, while label 1 blogs tend to have lower values, with some overlap and outliers. The blog-index axis is only an index and has no substantive interpretation. Since the plotted colors appear in long contiguous blocks, verify whether the dataset's original row order is grouped by political label; if so, randomize the plotting order or compare ratio distributions by label so the horizontal arrangement does not overstate separation. Labels are used only to color the plot after fitting, not as SCORE inputs.

Synthetic experiment design

The simulation generates undirected networks from a two-community DCBM with (n=500) nodes and

$$P = \begin{pmatrix} 0.08 & 0.02 \\ 0.02 & 0.08 \end{pmatrix}.$$

Within-community edges are therefore more likely than between-community edges. Node degree weights are generated from a lognormal distribution with dispersion fixed at $\sigma = 0.55$, matching the project's preliminary simulation. The analysis fits SCORE, oPCA, and nPCA to 30 independently generated networks. All three methods are evaluated on the same network within each replication using misclassification rate and ARI.

The generator caps extreme degree weights so that all Bernoulli probabilities remain at most 0.99. This safeguard slightly truncates the upper tail of the requested lognormal distribution. It samples each unordered node pair once and mirrors the edge to create an undirected network. The earlier R Markdown implementation sampled both directions independently before combining them, which changes the implied edge probabilities; results from that earlier generator are not equivalent to this corrected DCBM simulation.

In the illustrated network (seed 291), all three methods recover the two communities accurately. nPCA makes 3 errors (0.6%; ARI 0.976), SCORE makes 9 errors (1.8%; ARI 0.929), and oPCA makes 10 errors (2.0%; ARI 0.921). The adjacency matrix, ordered by true labels for display, shows denser within-community blocks than between-community edges. This pattern is consistent with the simulation design; because the network was generated with that block structure, it is not an independent validation of the method.

Across 30 networks at this fixed parameter setting, nPCA has the lowest mean misclassification rate (0.007, standard deviation 0.004) and highest mean ARI (0.972, standard deviation 0.016). oPCA has mean error 0.014 (SD 0.005) and mean ARI 0.944 (SD 0.019). SCORE has mean error 0.013 (SD 0.004) and mean ARI 0.947 (SD 0.017). Thus, all three methods recover these simulated communities well, with nPCA performing best on average in this particular design and SCORE close to oPCA.

The simulation ranking differs from the political blogs comparison, where SCORE performed best. This is not a contradiction: the simulated network follows one specific two-community DCBM with fixed block probabilities and a single degree-heterogeneity level. These results show that nPCA works especially well under this setting, not that it is generally superior or that SCORE fails. Method rankings may change as degree heterogeneity, sparsity, and community separation change.

The separate analysis file `21_RMD_PreliminarySims.Rmd` contains the simulation code and plots. For the 30-replication pilot, the reported summary is:

Method	Mean error rate	SD error rate	Mean ARI	SD ARI
nPCA	0.007	0.004	0.972	0.016
oPCA	0.014	0.005	0.944	0.019
SCORE	0.013	0.004	0.947	0.017

These values summarize the current analysis outputs. Rerun both analysis R Markdown files from a clean R session before final submission to regenerate and verify them.

Future Steps

Systematic simulation plan

The completed pilot uses one DCBM configuration: 500 nodes, two balanced communities, within-community probability 0.08, between-community probability 0.02, and lognormal degree heterogeneity $\sigma = 0.55$. The next simulation stage will vary degree heterogeneity, community separation, and overall sparsity while holding the other parameters fixed within each comparison. Additional runs can vary network size and the number and balance of communities. If time permits, a robustness experiment will add edge-observation errors or dependence to assess how methods behave outside the model's conditional-independence assumption. Each setting will use repeated networks, with SCORE, oPCA, and nPCA applied to the same network within each replication. I will report mean and standard deviation of permutation-adjusted misclassification rate and ARI, with comparative boxplots. These metrics directly evaluate recovery of categorical community labels.

The project partner's assigned method is SIMPLE, which tests whether a selected pair of nodes has the same membership profile and returns a test statistic and p-value. This differs from SCORE's task of assigning one community label to every node. I will compare them on shared simulated networks using task-appropriate measures: node-level error and ARI for SCORE, and type-I error, p-value calibration, and power for SIMPLE. On pure-membership simulations, I will also compare pairwise decisions: SCORE treats two nodes in the same estimated cluster as a same-community pair, while SIMPLE tests equality of the pair's membership profiles. If mixed-membership simulations are added, SIMPLE can be evaluated on profile equality; SCORE's hard labels do not estimate a full membership profile. Analyses involving many node pairs will account for multiple testing. Runtime and sensitivity to random initialization may be compared as secondary measures.

Real-data analysis

The political blogs network remains the primary application. It is available through the `blogosphere` data object in the R package `gsbm` and contains 1,222 blog nodes, 16,714 undirected hyperlinks, blog names, and liberal/conservative reference labels. I will verify node ordering across the adjacency matrix, names, and labels; check for missing or duplicate entries, self-loops, and isolated nodes; symmetrize the links; and document the degree distribution. Labels will remain outside the fitting step and be used only for evaluation and post-fit interpretation.

The analysis will examine which blogs are misclassified by SCORE, oPCA, and nPCA, and whether errors are associated with degree or with ambiguous positions in SCORE ratio space. The 50-seed nPCA check showed modest initialization sensitivity, with 590 to 600 errors; the main method comparison will use a fixed seed and report this check separately. For SIMPLE, I will specify a manageable set of node pairs whose membership-profile similarity is substantively meaningful, test those pairs using network-based estimates, and adjust for multiple comparisons if many pairs are tested. The political labels can provide an external benchmark, but they are not treated as error-free or as soft membership profiles.

Finally, I will rerun the real-data and simulation analyses from a clean R session, record package versions and random seeds, and include the generated tables and figures with the numbered function files and analysis R Markdown files. The final PDF should be knitted from the revised report source after results have been regenerated and checked.

References

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